

Sameer Paykel

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EDUCATION | California Polytechnic State University, San Luis Obispo (Cal Poly)

Bachelor of Science in Computer Science with Entrepreneurship Minor

June 2026

- GPA: 3.78 / 4.0 | *Magna cum laude* | Dean's List (10x) | President's List (2x)
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WORK EXPERIENCE

General Atomics Aeronautical Systems – Data Engineering Intern

June 2025 - August 2025

- Designed and deployed an Azure-based data pipeline (SQL, ETL) integrating legacy systems for UAV sustainment ops.
- Automated Python and SQL workflows, eliminating over 1,800 annual labor hours and enabling real-time reporting.
- Built ML-based document parsing system reducing technical lookup time by over 40%.
- Improved data reliability and query performance through schema design and pipeline optimization.

Renewable Energy Place (REplace) – Software Engineering Intern

June 2024 - August 2024

- Built and shipped full-stack features (React, Node.js, PostgreSQL) for a production renewable energy platform.
 - Designed geospatial data pipelines (GeoPandas, Mapbox) to process statewide energy datasets more efficiently.
 - Contributed to v2.0 platform launch enabling enterprise client onboarding.
 - Collaborated in Agile environment with code reviews, version control (Git), and sprint-based delivery.
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PROJECTS

Custom Programming Language – Racket

github.com/spaykel/zode

- Built a custom programming language including parser, interpreter, and runtime system.
- Implemented lexical scoping, evaluation strategies, and multiple execution models including typed/untyped and mutable/immutable language variants.
- Demonstrates strong foundation in compilers, systems, and language design.

Chess Artificial Intelligence Solver – Python w/ Tensorflow

github.com/spaykel/AI_ChessEngine

- Built a chess engine using Monte Carlo Tree Search (MCTS) with neural network evaluation.
- Optimized recursive search and pruning strategies to improve performance and reduce computational cost.

Event Finder Mobile App – React Native, Spring Boot, MySQL

- Built a full-stack mobile app for real-time, location-based event discovery.
 - Designed REST APIs and backend systems for user-generated content, moderation, and dynamic querying.
 - Implemented geolocation-based filtering and ranking for personalized user experience.
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LEADERSHIP

Hack4Impact Cal Poly – Director of Operations, Product Manager, & Developer

September 2023 - June 2026

- Led cross-functional teams (10+) delivering software solutions for nonprofit clients.
- Managed execution across 11 concurrent projects, including planning, prioritization, and delivery timelines.
- Oversaw budgeting, fundraising, and stakeholder coordination.
- Contributed to full-stack development and system improvements.

Sigma Phi Delta Fraternity – Treasurer

May 2025 - May 2026

- Manage a \$50,000+ annual budget as a member of the executive board for a 75+ member organization.
 - Oversee dues collection, financial planning, and transparent reporting to support chapter operations and decisions.
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SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL

Frameworks: React, Node.js, Spring Boot

Data/ML: TensorFlow, Pandas, NumPy

Tools: Git, REST APIs, PostgreSQL, Selenium

INTERESTS

Jazz Piano, DJing, Golf, Computer Building, Mandarin Chinese